

TAX ENGINE — FORMULAS & DATA REFERENCE

All calculation logic and parameter data for the ONYX Ledger engine · Tax years 2025 & 2026 · Prepared August 2026

How to use this: every formula here is deterministic — the engine is these formulas plus annual parameter tables. A companion file, `tax_engine_params.json`, holds the same numbers in machine-readable form so the code reads config, never hardcoded values. **Golden rule:** the 2025 figures apply to 2025 income (returns filed spring 2026); 2026 figures apply to income being earned now. Verify every number against CRA's published tables before certification season — this document is a build reference, not a certified source.

1. THE CORE PIPELINE — HOW EVERY RETURN COMPUTES

```
totalIncome      = employment + selfEmployment + interest + grossedUpDividends
                  + taxableCapitalGains + otherIncome           // T1 line 15000
netIncome        = totalIncome - deductions                     // line 23600
                  // deductions: RRSP, FHSA, childcare, union dues,
                  // moving, support paid, carrying charges
taxableIncome    = netIncome - additionalDeductions             // line 26000
                  // e.g. losses carried forward, security options deduction

fedTaxBeforeCredits = applyBrackets(taxableIncome, FED_BRACKETS[year])
provTaxBeforeCredits = applyBrackets(taxableIncome, PROV_BRACKETS[prov][year])

fedTax = max(0, fedTaxBeforeCredits - fedNonRefundableCredits - dividendTaxCredit_fed)
provTax = max(0, provTaxBeforeCredits - provNonRefundableCredits - dividendTaxCredit_prov
              + provincialSurtax)                                // ON only, see §4

totalPayable = fedTax + provTax + CPP_selfEmployed_owed
refundOrOwing = amountsPaid - totalPayable
               // amountsPaid: tax withheld on T4/T5 slips (box 22),
               // instalments paid, refundable credits (CWB, etc.)
               // positive = refund, negative = balance owing
```

```
function applyBrackets(income, brackets):           // brackets sorted ascending
    tax = 0
    for each (threshold, rate, nextThreshold) in brackets:
        slice = min(income, nextThreshold) - threshold
        if slice <= 0: break
        tax += slice * rate
    return tax
```

2. FEDERAL TAX BRACKETS

Rate-cut note: the lowest federal rate was cut from 15% to 14% effective July 1, 2025 — so tax year **2025 uses a blended 14.5%** on the first bracket, and **2026 uses 14%**. Non-refundable credits are converted to tax value at this same lowest rate (see §3), so the credit rate is 14.5% for 2025 and 14% for 2026.

Bracket (taxable income)	2025 rate	2026 bracket	2026 rate
\$0 – \$57,375	14.5%	\$0 – \$58,523	14%
\$57,375 – \$114,750	20.5%	\$58,523 – \$117,045	20.5%
\$114,750 – \$177,882	26%	\$117,045 – \$181,440	26%
\$177,882 – \$253,414	29%	\$181,440 – \$258,482	29%
over \$253,414	33%	over \$258,482	33%

Annual indexation: thresholds and most credit amounts are multiplied by CRA's indexation factor each year (2025: 2.7% · 2026: 2.0%). Store the factor; every November CRA publishes next year's numbers.

3. FEDERAL NON-REFUNDABLE CREDITS (the big ones)

All credits below are **amounts**; convert to tax reduction via: `creditValue = amount × lowestFedRate` (14.5% in 2025, 14% in 2026). Credits can reduce tax to zero but never below (except refundable ones, §9).

3.1 Basic Personal Amount (BPA) — with high-income phase-out

```
// 2025: max 16,129, min 14,538, phase-out range = 4th bracket (177,882 → 253,414)
// 2026: max 16,452, min ~14,829 (verify), range 181,440 → 258,482
if netIncome <= rangeStart:      bpa = bpaMax
else if netIncome >= rangeEnd:    bpa = bpaMin
else: bpa = bpaMax - (bpaMax - bpaMin) × (netIncome - rangeStart) / (rangeEnd - rangeSta
```

3.2 The rest of the standard stack

Credit	2025 amount	Formula / rule
Canada Employment Amount	\$1,471	<code>min(1471, employmentIncome)</code> — everyone with a T4 gets it
CPP/QPP contributions	actual paid	base CPP contributions are a credit; the enhanced portion (incl. CPP2) is a deduction from income instead — see §5
EI premiums	actual paid (max \$1,077.48)	credit for premiums from T4 box 18
Age amount (65+)	\$9,028	reduced by 15% of <code>netIncome</code> over \$45,522, to zero

Medical expenses	threshold-based	$\max(0, \text{eligibleMedical} - \min(0.03 \times \text{netIncome}, 2834))$ — claim on either spouse; the lower-income spouse usually wins (lower 3% threshold). 12-month claim window ending in tax year
Charitable donations	two-tier	$\text{lowestRate} \times \min(200, \text{don}) + 29\% \times \text{amountOver200}$ (33% on the portion matching income in the 33% bracket). Combine spouses; can carry forward 5 years — optimizer opportunity: pooling
Tuition (T2202)	actual	credit at lowest rate; unused amounts carry forward indefinitely or transfer up to \$5,000 to parent/spouse
Spouse / eligible dependant	= BPA	BPA-sized amount reduced dollar-for-dollar by the dependant's net income
Disability (DTC)	\$10,138	requires CRA-approved T2201; transferable from dependant
Home buyers' amount	\$10,000	flat, year of first-home purchase
First \$200 / over \$200 donation rates	first-tier uses lowest rate (14.5%/14%); keep as parameters, they moved with the 2025 rate cut	

4. PROVINCIAL TAX (2025 brackets — index annually)

Same applyBrackets function, different tables. Each province also has its own BPA and credit set (store per-province params). Quebec excluded (separate TP1 system, out of scope at launch).

Province	2025 brackets → rates	2025 prov. BPA
Ontario	to \$52,886: 5.05% · to \$105,775: 9.15% · to \$150,000: 11.16% · to \$220,000: 12.16% · above: 13.16%	\$12,747
British Columbia	to \$49,279: 5.06% · to \$98,560: 7.7% · to \$113,158: 10.5% · to \$137,407: 12.29% · to \$186,306: 14.7% · to \$259,829: 16.8% · above: 20.5%	\$12,932
Alberta	new for 2025: to \$60,000: 8% · to \$151,234: 10% · to \$181,481: 12% · to \$241,974: 13% · to \$362,961: 14% · above: 15%	\$22,323
Saskatchewan	to \$53,463: 10.5% · to \$152,750: 12.5% · above: 14.5%	\$19,491
Manitoba	to \$47,564: 10.8% · to \$101,200: 12.75% · above: 17.4%	\$15,780
Nova Scotia	to \$30,507: 8.79% · to \$61,015: 14.95% · to \$95,883: 16.67% · to \$154,650: 17.5% · above: 21%	\$11,744
New Brunswick	to \$51,306: 9.4% · to \$102,614: 14% · to \$190,060: 16% · above: 19.5%	\$13,396
PEI	to \$33,328: 9.5% · to \$64,656: 13.47% · to \$105,000: 16.6% · to \$140,000: 17.62% · above: 19%	\$14,250
Newfoundland & Labrador	8 brackets, 8.7% (to \$44,192) rising to 21.8% (over ~\$1.1M)	\$11,067

Verify all provincial figures against each province's published 2025/2026 tables before shipping — provinces index by their own factors (some, like PEI historically, don't index every amount), and Nova Scotia only began indexing in 2025. Treat this table as scaffolding.

4.1 Ontario surtax — the one weird extra formula

```
// applied AFTER Ontario credits, on Ontario basic tax:
surtax = 0.20 × max(0, onBasicTax - 5710) + 0.36 × max(0, onBasicTax - 7307) // 2025 t
onTax  = onBasicTax + surtax - ontarioTaxReduction // low-income reduction also exists
```

5. PAYROLL DATA — CPP, CPP2, EI (needed for credits, self-employed, and refund accuracy)

Parameter	2025	2026
CPP earnings ceiling (YMPE)	\$71,300	\$74,600

CPP basic exemption	\$3,500	\$3,500
CPP rate (employee)	5.95%	5.95%
CPP max (employee)	\$4,034.10	\$4,230.45
CPP2 second ceiling (YAMPE)	\$81,200	\$85,000
CPP2 rate / max (employee)	4% / \$396.00	4% / \$416.00
EI insurable max (MIE)	\$65,700	\$68,900
EI rate / max (employee)	1.64% / \$1,077.48	1.63% / \$1,123.07

```
cpp = 0.0595 × clamp(pensionableEarnings - 3500, 0, YMPE - 3500)
cpp2 = 0.04 × clamp(pensionableEarnings - YMPE, 0, YAMPE - YMPE)
ei = eiRate × min(insurableEarnings, MIE)
// self-employed pay BOTH halves of CPP (11.9% / 8%); half is a deduction, base portion
// credit/deduction split: base CPP (4.95% part) → credit; enhanced (1% part + CPP2) → d
```

6. INVESTMENT INCOME FORMULAS

6.1 Capital gains

```
taxableCapitalGain = 0.50 × (proceeds - adjustedCostBase - sellingCosts)
// inclusion rate is 50% – the proposed 2/3 increase was CANCELLED; do not build it in
// capital LOSSES offset gains only; net losses carry back 3 yrs / forward indefinitely
// superficial loss rule: loss denied if same security bought within ±30 days
```

6.2 Dividends (Canadian) — gross-up and credit

```
// ELIGIBLE dividends (most public companies, T5 box 24/25):
grossedUp    = dividend × 1.38
fedDTC       = grossedUp × 0.150198
// NON-ELIGIBLE (small business, box 10/11):
grossedUp    = dividend × 1.15
fedDTC       = grossedUp × 0.090301
// provinces have their own DTC rates – parameter table per province
// interest & foreign dividends: no gross-up, fully taxable at marginal rate
```

7. REGISTERED ACCOUNTS — LIMITS AND RULES

Account	2025	2026	Rules the engine must enforce
RRSP	18% of prior-year earned income, max \$32,490	max \$33,810	room accumulates (carry-forward from CRA Notice of Assessment); contributions in first 60 days of a year deductible against prior year; over-contribution buffer \$2,000 then 1%/month penalty; deduction can be deferred to a future higher-income year (optimizer opportunity)
TFSA	\$7,000	\$7,000	no deduction, growth tax-free; room accumulates from age 18/2009; withdrawals restore room next Jan 1 (re-contributing same year = over-contribution, 1%/mo)
FHSA	\$8,000/yr	\$8,000/yr	\$40,000 lifetime; deduction like RRSP and tax-free withdrawal for first home (best of both); unused annual room carries forward max \$8,000; 15-year max life; must be first-time buyer to open
RESP	\$50,000 lifetime/child		CESG grant = 20% of first \$2,500/yr = \$500 free money (optimizer nudge)

8. REFUND ESTIMATION (the number on your dashboard)

```
estimatedRefund = taxWithheld // Σ T4 box 22, T5 withholding
                  + instalmentsPaid
                  + refundableCredits // CWB, eligible educator, etc.
```

```
        - totalPayable                                // from §1
// CONFIDENCE: show a range, not a point. Drivers of uncertainty:
//   missing slips (AFR pull incomplete), unentered deductions, pending RRSP (first-60-d
// display: "Estimated refund $X - $Y" until all declared sources are entered
```

Why withholding $\neq$ tax owed (and refunds exist): employers withhold per-paycheque as if that pay rate ran all year, ignoring RRSP contributions, medical, donations, and multi-job effects. Two jobs = each withholds as if it's the only income = usually a balance **owing**. Detect the multiple-T4 case and warn the user early — it's the #1 "surprise bill" scenario.

9. MARGINAL vs EFFECTIVE RATE (dashboard metrics)

```
marginalRate = fedRate(taxableIncome) + provRate(taxableIncome)    // rate on the NEX
effectiveRate = (fedTax + provTax) / totalIncome                    // average on ALL
// example, ON 2025 @ $80,000 taxable: marginal = 20.5% + 9.15% = 29.65%
// note for accuracy: marginal rate near credit phase-out zones (BPA clawback, age amoun
// OAS clawback) is HIGHER than the bracket rate – compute marginal numerically:
trueMarginal = tax(income + 100) - tax(income) / 100              // robust method,
```

10. SIMULATOR FORMULAS (your differentiator)

10.1 RRSP contribution simulator

```
// NEVER use flat "contribution × marginalRate" – wrong when crossing brackets. Do this:
taxSaving = totalTax(taxableIncome) - totalTax(taxableIncome - contribution)
// where totalTax = fed + prov, full pipeline both times. Also recompute income-tested
// benefits (CCB, GST credit, OAS clawback) – RRSP deductions can INCREASE benefits;
// showing that combined number is a feature competitors don't have
alsoShow: refundPerDollar curve, room remaining, over-contribution warning
```

10.2 FHSA simulator

```
same taxSaving formula as RRSP (it's a deduction), capped at min(8000 + carryforward, 40
comparison view: FHSA vs RRSP-HBP vs TFSA for a down-payment goal
```

10.3 Capital gains timing simulator

```
taxThisYear = totalTax(income + 0.5 × gain) - totalTax(income)
taxNextYear = totalTax(projectedNextIncome + 0.5 × gain) - totalTax(projectedNextIncome)
show: difference, plus December/January split option (half the gain in each year
      uses two sets of brackets – legitimate and material for large gains)
```

10.4 Donation pooling simulator

```
// carrying forward 5 yrs & combining spouses beats claiming $150/yr repeatedly,
// because only amounts OVER $200/yr earn the 29% rate:
annualClaim = 5 × (lowestRate × 150) // five small claims
pooledClaim = lowestRate × 200 + 0.29 × (750 - 200) // one pooled claim
```

11. TAX HEALTH SCORE — PROPOSED FORMULA (your own metric — you define it)

```
score = Σ weightedComponents, scaled to 0-100:
  25 pts registeredAccountUsage = 25 × usedRoom / availableRoom (RRSP+FHSA+TFSA ble
  20 pts deductionCapture       = 20 × claimedDeductions / detectedEligible
    (medical, donations, union dues, childcare, WFH-detailed, tuition)
  15 pts withholdingAccuracy    = 15 × (1 - |refundOrOwing| / totalPayable), floor 0
```


(big refund = interest-free loan to CRA; big owing = planning failure — both s

15 pts filingHygiene = on-time history, no missing slips vs AFR, instalmen

15 pts incomeStructure = credit for tax-efficient asset location

(interest inside registered accounts, dividends/gains outside, income splittin

10 pts planningEngagement = simulators run, RRSP deadline met, carry-forwards t

// Every point lost MUST map to a concrete action card ("Contribute \$X to RRSP: +N pts,

// — the score is only defensible if its dollar figures come from THIS engine, not marke

12. INCOME-TESTED CLAWBACKS (needed for accurate marginal rates)

Program	2025 parameters	Formula
OAS recovery tax	threshold $\approx$ \$93,454 (verify)	clawback = $0.15 \times \max(0, \text{netIncome} - \text{threshold})$, to 100% of OAS
Canada Child Benefit	base \$7,997/child <6, \$6,748 ages 6–17 (2025-26 year)	reduced by family-income tiers — implement CRA's published phase-out table; RRSP deductions raise CCB (simulator hook)
GST/HST credit & CWB	CRA tables	refundable; phase in/out by family net income — table-driven, revisit annually

13. THINGS YOUR DEMO IMPLIES THAT ARE WRONG OR DEAD — FIX BEFORE CODING

- **Work-from-home \$2/day flat method is DEAD** (ended after 2022). WFH now requires the **detailed method**: employer-signed T2200 + T777 form, claiming workspaceArea/homeArea $\times$ eligibleHomeCosts (rent, utilities; commission employees add more; **mortgage principal never eligible**). The demo's "+\$180 WFH" card must be built on this, not the flat rate.
- **Consumer carbon rebate (CCR) ended April 2025** — no rebate card in the product.
- **Capital gains inclusion stays 50%** — the announced 2/3 increase was cancelled; don't build remnants of it.
- **The lowest federal rate is 14.5%/14%, not 15%** — any calculator copied from a pre-2025 tutorial will be wrong on every bracket-1 dollar and every credit.

14. HOW TO STRUCTURE THIS IN CODE

- **Everything above is config, not code.** One tax_engine_params.json keyed by year $\rightarrow$ jurisdiction $\rightarrow$ parameter (companion file provided). The formulas never change annually; only the tables do. Annual update = new JSON block + rerun test suite.
- **Money math:** compute in cents (integers) or a decimal type — never floating point. CRA rounding: per-line rounding to the cent; final balances to the dollar follow form rules.
- **Test harness from day one:** build $\sim$ 50 personas (single T4; two T4s; T4+RRSP; self-employed; investor with gains/dividends; senior with OAS; parent with CCB; each province) and lock expected outputs. When

CRA's certification test returns arrive, they slot into the same harness. Cross-check personas against an existing certified product's results while developing.

- **Every computed number carries provenance:** store which formula + which parameters produced it, so the UI can show "why" (trust feature) and support can debug (liability feature).

Prepared August 2026 from CRA-published 2025 figures and announced 2026 indexation (factor 2.0%). Provincial tables, clawback thresholds marked "verify," and all credit amounts must be re-validated against official CRA / provincial publications each November before the filing season. This document is an engineering reference, not tax advice.